Codes for importing the necessary libraries:

```
import numpy as np
import matplotlib.pyplot as plt
from numpy.polynomial import Polynomial
```

Question 01 [4 Marks]

Represent the following polynomial using the **Polynomial class of the Numpy library**.

Let, $g(x) = 3x^5 - 2x^4 + 4x^2 - 5x + 7$

A. **[1 Mark]** Find $4 \times g(-2) - 6$.

B. **[1.5 Marks]** Find out the value of $(3g'''(x) - g'(x))\,/\,2$ at $x = 2$.

C. **[1.5 Marks]** Plot $g(x)$ with a **blue line** and $g''(x)$ (the second derivative) with a **red line** in the same graph $-3 \leq x \leq 4$.

Question 02 [8 Marks]

A temperature-monitoring system records temperature values and their corresponding rates of change at several time points. The recorded data points are represented as: (x_0, y_0), $(x_1 y_1)$, (x_2, y_2)....... (x_n, y_n)

Let, data_X = $[x_0, x_1, x_2, ..., x_n]$ represents the measurement times.

and data_Y = $[y_0, y_1, y_2, ..., y_n]$ represents the recorded temperatures.

To construct a smooth approximation of the temperature behaviour, an engineer defines the following **modified Lagrange Basis**:

$$l_k(x) = \prod_{j=0, j \neq k}^{n} \frac{x_k - x_j}{x - x_j}$$

Using the modified Lagrange basis, the following **modified Hermite bases** are defined:

$$h_k(x) = (2x + 4(x^3 - x_k)l_k^2(x))$$

$$\hat{h}_k(x) = (x^2 - x_k)l_k(x) - x + 3x^3$$

A. [5 Mark] Define functions to calculate the modified Lagrange basis $l_k(x)$ and the modified Hermite bases $h_k(x)$ and $\hat{h}_k(x)$.

B. [3 Mark] Find the **values** of the Lagrange basis and both Hermite bases when k = 2 and $x = 2$.

Sample Input	Sample Output
data_X = np.array([5, 13, 22, 80]) data_Y = np.array([5, -75, -50, -20]) k = 2 x = 2	l_2(x) = 3.447552447552448 h_2(x) = -661.594601202993 h_hat2(x) = -40.05594405594406

Question 03 [8 Marks]

Suppose there is a function,

$$f(x) = 3x^5 - 3x^4 + 7x^2 + 3x + 6$$

To analyze the behavior of the Absolute Error when approximating the derivative of the function at $x = 3$ using **Forward Difference**, you have to determine the minimum value of h that ensures the absolute error stays below a given tolerance threshold of 0.002463. Additionally, calculate how many reductions of h are required to meet this condition.

As you already know,

A. Forward Difference, $f'(x) = \frac{f(x+h)-f(x)}{h}$

B. Absolute error = |Actual derivative - Approximate derivative|

You should start with an initial step size of ***h* = 1.5** and iteratively reduce h by one-third, if necessary, until the error condition is satisfied.

Hint: You have to use the Polynomial class from the numpy library to define the function, $f(x)$ and your output should clearly state the **minimum *h* *[4 Marks]*** and the **total number of reductions *[4 Marks]*** required to achieve the desired accuracy.

Another Hint: In each iteration, update h by dividing it by 3 (i.e., h = h / 3). For example, if h = 1.5 in the current iteration, then in the next iteration h = 1.5 / 3 ≈ 0.5.

YOUR CODE SHOULD GIVE OUTPUT
Minimum h: $2.8225146347383815 \times 10^{-06}$ Total number of reductions: 12